

COMS5023A: Task 3-File upload and Table Join

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Total Marks: 50



Student_Records.xlsx

Question 1 (4 Marks) - Importing Data and Table Creation

Create a database named WitsStudents. Import the spreadsheet Wits_Student_Records.xlsx into SQL Server/MySQL. Rename the imported table to StudentRecord. Ensure suitable data types are assigned to each column (e.g., StudentID INT, DateOfBirth DATE, YearOfStudy INT, and text fields VARCHAR). Take a screenshot showing the table structure/design view with all column names and data types and paste it into your submission.

Question 2 (1 Mark)

Display all records from the StudentRecord table.

Question 3 (3 Marks)

Retrieve StudentID, FirstName, LastName and DateOfBirth for students born after 01 January 2005.

Question 4 (3 Marks)

Display all students who are in YearOfStudy 2. Sort the results by LastName in ascending order.

Question 5 (3 Marks)

Use a logical operator to display students registered for BSc Data Science AND who are in YearOfStudy 1.

Question 6 (3 Marks)

Use logical operators to display students who are in YearOfStudy 3 OR YearOfStudy 4.

Question 7 (2 Marks)

Use the LIKE wildcard operator to find all students whose LastName starts with the letter M.

Question 8 (2 Marks)

Using SQL functions, display each student's age in years based on the DateOfBirth field.

Question 9 (3 Marks)

Count the number of students registered for each CourseRegistered. Display the course name and total number of students.

Question 10 (2 Marks)

Using GROUP BY and HAVING, display only courses that have more than 5 students registered.

Question 11 (2 Marks)

Display the number of students per YearOfStudy. Sort the results from the highest number of students to the lowest using ORDER BY DESC.

Question 12: Create and Populate the Tables (5 Marks)

Create the following tables in a database named WitsStudents and insert the sample records provided.

Table 1: Students

StudentID	FirstName	LastName	YearOfStudy
2026001	Nokwanda	Mashaba	1
2026002	Lerato	Mokoena	2
2026003	Tsakani	Baloyi	1
2026004	Rendani	Netshidzivhani	2
2026005	Aphiwe	Mgangatho	3

Table 2: Registrations

RegistrationID	StudentID	CourseRegistered	Faculty
R001	2026001	BSc Data Science	Science



R002	2026002	BSc Computer Science	Science
R003	2026003	BSc Data Science	Science
R004	2026006	BSc Mathematics	Science
R005	2026007	BSc Statistics	Science

Question 13: INNER JOIN (2 Marks)

Write an INNER JOIN query to display StudentID, FirstName, LastName and CourseRegistered for students that have matching records in both tables.

Question 14: INNER JOIN with ORDER BY (2 Marks)

Display all matching students and courses using an INNER JOIN. Sort the results by LastName in ascending order.

Question 15: LEFT JOIN (2 Marks)

Write a LEFT JOIN query to display all students from the Students table, including those without matching registration records.

Question 16: LEFT JOIN with NULL Filter (3 Marks)

Using a LEFT JOIN, display only students who do not have a matching registration record.

Question 17: RIGHT JOIN (2 Marks)

Write a RIGHT JOIN query to display all records from the Registrations table, including registrations that do not have matching students.

Question 18: RIGHT JOIN with Course Filter (3 Marks)

Use a RIGHT JOIN to display registrations for BSc Data Science and the corresponding student details where available.



Question 19: FULL JOIN (2 Marks)

Write a FULL OUTER JOIN query to display all records from both tables, regardless of whether a match exists.

Question 20: FULL JOIN Analysis (2 Marks)

Using a FULL OUTER JOIN, identify records that exist in one table but not the other.

Question 21: JOIN with Aggregate Function (2 Marks)

Using an INNER JOIN, display the number of registered students per course. Use GROUP BY.

Question 22: JOIN with HAVING Clause (3 Marks)

Using an INNER JOIN, display courses that have more than one registered student using GROUP BY and HAVING.